

BIOL 2021: Environmental Issues Lab

CRN: 24017 (-01); 24244 (-02)

1 CREDIT HOUR

COURSE DATES: Fall 2025

Thursday 12 – 2:50 PM, Meyer 125 (-01)*

Thursday 8 – 10:50 AM, Meyer 125 (-02)*

*Many class meetings will involve field trips. There is also a weekend field trip to the ONU Metzger Nature Center (Nov. 14 - 16) and a full-day field trip to the Water Management Association of Ohio conference (Thursday, Nov. 13).

INSTRUCTORS

Dr. Leslie Riley

Office Hours: MW 11 AM – 1 PM; F 11AM – 12 PM and by appointment

Office Location: Meyer 118

Telephone: (419) 772 – 3143

E-mail: l-riley.1@onu.edu

Dr. Liz Tristano

Office Hours: MWF 9 AM – 11 AM and by appointment

Office Location: Meyer 154

Telephone: (419) 772 – 2327

Email: e-tristano@onu.edu

CATALOG DESCRIPTION

This course provides hands-on experiences to complement the concepts covered in Environmental Issues (BIOL 2011). As an inherently interdisciplinary field, activities and trips are designed to explore the physical, chemical, biological and earth sciences to further understand how the environment works, how humans interact with the environment, and how possible solutions are implemented to address environmental problems. There is an additional fee for this course.

COURSE COREQUISITE(S)

BIOL 2011

GENERAL EDUCATION

N/A

COURSE READINGS AND MATERIALS

- Rite-in-the-Rain notebook (universal pocket-top spiral) – instructors will provide
 - Backpack, water bottle, hiking boots or shoes, raincoat
 - Course readings and assignments will be available on Canvas
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COURSE OUTCOMES

This course uses firsthand observation and a hands-on approach to develop a deeper understanding of the concepts covered in Environmental Issues (Biol 2011). At the conclusion of the semester, students should be able to:

- Evaluate environmental issues through the application of scientific and information literacy
- Investigate the natural world using the fields of environmental science, ecology and evolution
- Explore how human population size and resource use can affect environmental impacts (footprints)
- Demonstrate how anthropogenic activities can affect ecology, biodiversity and natural resources (e.g. land, water, food, energy and air).
- Propose, evaluate and debate solutions to complex environmental problems

PROFESSIONAL SKILLS

During this course, you will also be exposed to a variety of skills that could be useful in your future professional careers. Some of the skills that you develop during your time here might also be useful to include on a resume. These include, but are not limited to the following:

- Stream assessment techniques
- Logging field observations
- Data analysis and graphing in Excel
- Restoration management techniques

COURSE POLICIES

Attendance: Required. Excused absences must be approved at least one week prior to the scheduled absence. If you are sick, please let us know 24 hours in advance so we can make alternative arrangements. Due to the hands-on nature of this course, more than TWO unexcused absences will result in a failing grade for the course.

Evaluation: Your final grade will be based upon the following criteria:

• Field Notebook	20
• Lab Write-Ups & Activities (14 @ 10 points each)	140
• Final Exam	20

Total Points	180
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Grading Scale: A: 90 – 100%, B: 80 – 89%, C: 70 -79%, D: 60 – 69%, F: 59% and below

Late Work: Late work will be accepted with a deduction of 10% per day, up to one calendar week after the due date.

Field Trips: During many class meetings there will be field trips to off campus locations to meet with local experts and explore various environmental issues. We expect everyone to be respectful of our hosts and their facilities. Failure to do so will result in field trip privileges being revoked and a failing grade for the course.

Weekend Field Trip & Conference Attendance: We will have one weekend field trip to the Metzger Nature Center (Tuscarawas County, Ohio) as well as a full-day field trip to the Water Management

Association of Ohio Annual Conference. If you have conflicts with these dates, please let us know ASAP. More details will be forthcoming about both trips.

Other Expectations: Please come to lab on time and prepared for the day's activities. This will often include reading material ahead of time or completing online virtual tours. This also includes having a copy of the lab handout with you (if applicable) as well as your field notebook and pencil.

TENTATIVE COURSE SCHEDULE

Week	Date	Topic	Location
Week 1	28-Aug	Environmental Hot Topics	ONU Campus
Week 2	4-Sep	Environmental Observations	Tidd-Oakes Farm
Week 3	11-Sep	Environmental Pollution	ONU Campus
Week 4	18-Sep	Sustainability Trends	ONU Campus
Week 5	25-Sep	Invasive Species	Haver Ridge Park
Week 6	2-Oct	Stream Assessment	Ottawa River Watershed
Week 7	9-Oct	Organic Agriculture	Little Riley Creek Farm
Week 8	16-Oct	Agriculture & Water Quality	Kurt Farm, Blanchard River Watershed
Week 9	23-Oct	Drinking Water Treatment	Lima Drinking Water Treatment Plant
Week 10	30-Oct	Solid Waste Management	Wyandot Recycling Center & Landfill
Week 11	6-Nov	Restoration Management	Johnny Appleseed Park District
Week 12	13-Nov	WMAO Conference	Columbus, Ohio
	Nov. 14 - Nov. 16	Metzger Nature Center Weekend Trip	ONU Metzger Nature Center
Week 13	20-Nov	Wastewater Management	Lima Wastewater Treatment Plant
	Nov. 24 - Nov. 28	Thanksgiving Break	
Week 14	4-Dec	Restoration Management	Tidd-Oakes Farm



Week 15	11-Dec	Wrap-Up & Final Exam	ONU Campus
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COURSE FEES

\$50; Nature Center: \$105

ONU UNIVERSAL COURSE POLICIES

Ohio Northern University is dedicated to providing an equitable educational experience for all enrolled students. Universal course policies applicable to all courses can be found at the following link: https://my.onu.edu/registrars_office/policies. Specifically, this website includes the policies for the following topics:

- Academic Dishonesty Policy
 - Academic Accommodations Policy
 - ONU Mercy Health Center Policy
 - Title IX Policy
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